

SARA-Net: A Multimodal Deep Learning Framework for Flood Risk Assessment in the Sundarbans using SAR Imagery

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Abstract—The Sundarbans and surrounding regions are increasingly threatened by climate change, which often leads to severe flooding and cyclone damage. Accurate predictions of these events are essential for protecting local communities and ecosystems. This paper proposes SARA-Net, a deep learning model that predicts flood risks by integrating satellite radar imagery with statistical climate data. The method uses a dual-stream architecture that concurrently analyzes environmental records and visual terrain features. The model was trained and tested on a dataset from 2014 to 2025, covering the districts of Satkhira, Khulna, and Bagerhat. A classification accuracy of 83.71% and an AUC of 0.9349 were achieved during testing and validation, indicating a strong capability to identify high-risk scenarios. Feature importance analysis revealed that satellite image data is the main factor influencing prediction accuracy. Additionally, a long-term risk analysis accurately identified significant historical events, including Cyclone Amphan in 2020 and peak flood risks in 2022. Explainable AI techniques were used to confirm that the model correctly focuses on riverbanks and flooded areas rather than irrelevant background noise. This research provides a reliable, validated tool for assessing flood vulnerability in the Sundarbans.

Index Terms—Deep Learning, Explainable AI, Flood Risk Assessment, Multimodal Fusion, Remote Sensing, Synthetic Aperture Radar (SAR).

INTRODUCTION

The Sundarbans is an important ecosystem shared by Bangladesh and India. It is the largest mangrove forest in the world and serves as a shield for millions of people living in the deltaic region. This forest protects the land from frequent and intense cyclones in the Bay of Bengal. Beyond

the protection, the Sundarbans support the local economy by helping communities that depend on fishing, honey collection, and wood gathering. However, this natural defense is facing a severe crisis. Local human needs and global climate change are messing up the balance between fresh river water and salty sea water. Recent studies show that forests are declining rapidly due to rising sea levels, changing rainfall patterns, and salt intrusion into the soil [9], [12].

The Sundarbans' problems are complex, including physical changes such as riverbank erosion and thinning trees. However, an invisible problem is making it difficult for mangrove plants and crops to grow, as soil chemistry is changing due to salty. This environmental damage harms the people living there as well. Many social scientists have done surveys to understand how climate change affects people. Their studies help identify long-term gaps that locals may not realize they face [7], [15]. These social studies cannot explain what is happening across the entire ecosystem because the monitoring system is weak and insufficient, focusing only on small areas and specific time periods, even though these concerns are severe. On the other hand, researchers have mapped the green cover loss over the past few decades using satellite data from Landsat and Sentinel-2 [6], [8]. However, relying only on these optical images is a big problem in a developing country like Bangladesh. During monsoon season, heavy clouds cover the sky for extended periods, leaving optical satellites blind when they need them most to monitor floods and disasters. It creates significant gaps in the data and stops them from tracking

extreme weather in real time [16]. To actually understand the Sundarbans, we need to look at the forest and the climate data together. Previous studies have commonly analyzed salinity or forest loss separately, without combining them into a single model [4], [13].

In this research, a new solution to these problems is proposed using a Multimodal Data Fusion system. Instead of relying on a single data source, we combined two data sources to create a comprehensive model. First, we used Synthetic Aperture Radar (SAR) images from Sentinel-1 for the years 2014 to 2025. Unlike standard optical methods, which are affected by clouds, SAR sensors use microwave pulses to see through clouds and deliver reliable data during monsoon seasons. Second, we merged this image data with actual text-based climate data collected over the same specific time. We specifically selected three key districts. By dividing the dataset, we can analyze the effects of climate change more accurately. We used Artificial Intelligence (AI) to process this combined data. A study reports that deep learning models that handle multiple data types perform better than traditional methods [14]. Our system seamlessly integrates diverse data sources, enabling adoption of this technology without extensive technical expertise. This integration enhances the system's usability and scalability for practical environmental monitoring and decision making.

We used this approach to achieve three distinct objectives: first, to identify the exact locations where "worst-case" flood scenarios are most likely to occur; second, to use reflexive heatmaps to visualize the overall geographic impact of these disasters; and third, to deliver an AI-driven visualization of actual flood occurrences based on the 12-year trend analysis (2014-2025). This research makes a significant contribution to the field by demonstrating that combining radar images with climate data can overcome cloud-coverage issues in tropical regions. In practice, it provides weatherlogists a powerful tool to monitor the Sundarbans proactively. Transitioning from simple observation to risk prediction allows conservationists and authorities to implement preventive measures before disasters strike, ultimately safeguarding lives and protecting ecosystems.

RELATED WORK

Recent literature on the Sundarbans highlights the critical need for improved environmental monitoring and information. Recognizing the human aspect of climate change, Islam [1] conducted studies in peripheral coastal villages and found that, although 90% of residents face environmental threats, 79% lack the resources to adapt. Similarly, Chakma et al. [3] conducted a survey in the Satkhira District and discovered that saline intrusion has forced approximately 30% of respondents to change professions, often moving away from traditional resource collection. A lack of future planning was also noted by Islam et al. [15], who observed that many villagers remain "unconcerned" due to limited resources and knowledge. According to Geary and Scheffran [5], technocratic approaches often worsen inequality. On the other hand, Siddique et al.

[7] observed that residents are often excluded from decision-making, despite policies that promote community cooperation. Although these studies offer important context, they are primarily limited to small sample sizes and specific regions, making it challenging to generalize their findings to the entire ecosystem.

Simultaneously, satellite imagery has become the standard for monitoring regional physical changes. Enan et al. [2] tracked forest degradation using Landsat imagery and found that cyclones were the leading cause of forest loss. On the other hand, Roy et al. [6] and Hossain et al. [8] mapped the decline in dense forest cover across multiple decades using Sentinel-2 and historical data. Wahid et al. [12], and Nandi et al. [19] focused on water quality and analyzed long-term salinity trends, confirming sharp increases in the central Sundarbans. However, there remains a significant limitation in these optical studies: they are dependent on weather conditions. According to Shampa et al. [16] and Sowrav et al. [11], cloud cover makes optical satellites useless during the monsoon season. While some researchers have sought to increase accuracy by combining optical and SAR data, as in Tusar et al. [4] and Toosi et al. [22], aligning these datasets remains a challenging technical task.

To overcome these limitations, deep learning is rapidly transforming how environmental data is analyzed. Li et al. [14] demonstrated that models like CNNs enable the successful integration of various data sources. In the context of flood monitoring, Rahman et al. [20] showed that traditional statistical methods significantly outperform hybrid models. Behera et al. [21] created tools to assist farmers in predicting salinity for irrigation, whereas Wang et al. [23] used computer vision to map mangroves with high precision. The "black box" problem in AI remains a significant challenge despite these developments. According to flood-mapping research by Chakma and Akter [17] and Dewan et al. [18], while AI may recognize flooded areas, it rarely explains *why* a specific forecast was made. Moreover, few studies have effectively combined tabular climate text data with unprocessed satellite imagery to create a coherent forecasting system.

Our work addresses two particular gaps identified in the literature. First, we used SAR data for all-weather monitoring, in contrast to optical-based studies that struggle with clouds [16]. Second, as mentioned by Rahman et al. [20], we address the transparency limitation by implementing the LayerCAM visualization technique [24], which provides a clear understanding of which geographic features influence the AI's risk assessment.

METHODOLOGY

Our study presents a Multimodal Data Fusion framework to predict flood risk in the Sundarbans by integrating satellite imagery with environmental text data. The whole workflow of our system is illustrated in the figure 1.

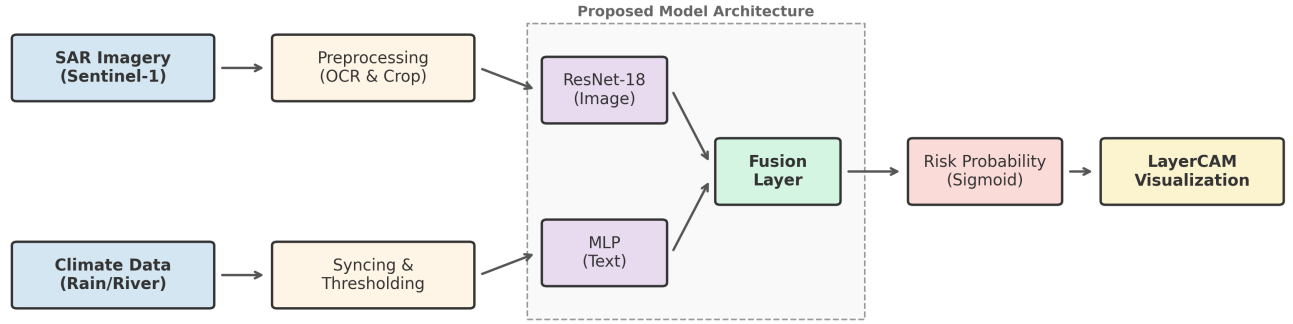


Fig. 1. The complete workflow of the proposed SARA-Net system.

A. Dataset Description

We assembled a multimodal dataset covering the years 2014–2025 using two primary sources:

- 1) **Visual Data (SAR):** Sentinel-1 imagery "SAR image of Sunderbans" dataset [25] was used for its all-weather capability. The dataset has three distinct processing levels:
 - *Enhanced Visualization:* RGB composites are used for qualitative analysis of significant events such as cyclones.
 - *SAR Urban:* Designed for effectively monitoring infrastructure, embankments, and human encroachment.
 - *VH Polarization:* Sensitive to the structure of vegetation, supports the monitoring of mangrove health and biomass.
- 2) **Textual Data (Climate):** Sourced from the "Bangladesh Climate Change Simulation Dataset" [26], we extracted key flood drivers: Annual Rainfall (mm) and River Water Levels (m).

B. Data Preprocessing and Labeling

To prepare our dataset, we created an automated procedure. To associate each image with its corresponding year, we first used Optical Character Recognition (OCR) to extract timestamps from the raw SAR images. Then, the satellite photos were divided into three distinct zones representing Satkhira, Khulna, and Bagerhat. It helps AI to analyze flood risks for each specific zone rather than the entire region at once. Lastly, we merged each image with its corresponding rainfall and river-level data from the second dataset. The median Flood Impact Score (4.69) was used to classify the data. Images with a score above the line were labeled "High Risk" (Class 1), while those below the line were labeled "Safe" (Class 0).

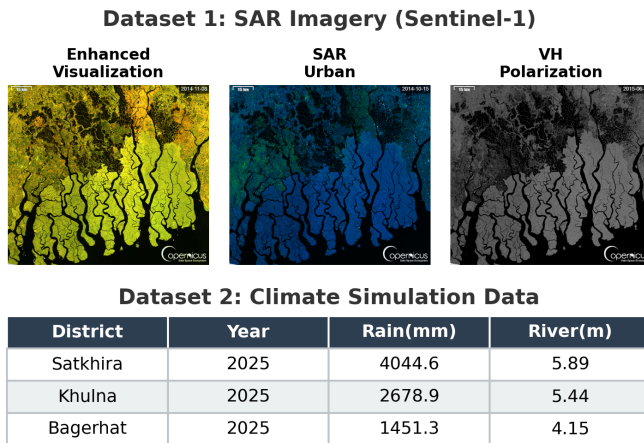


Fig. 2. Overview of the multimodal dataset.

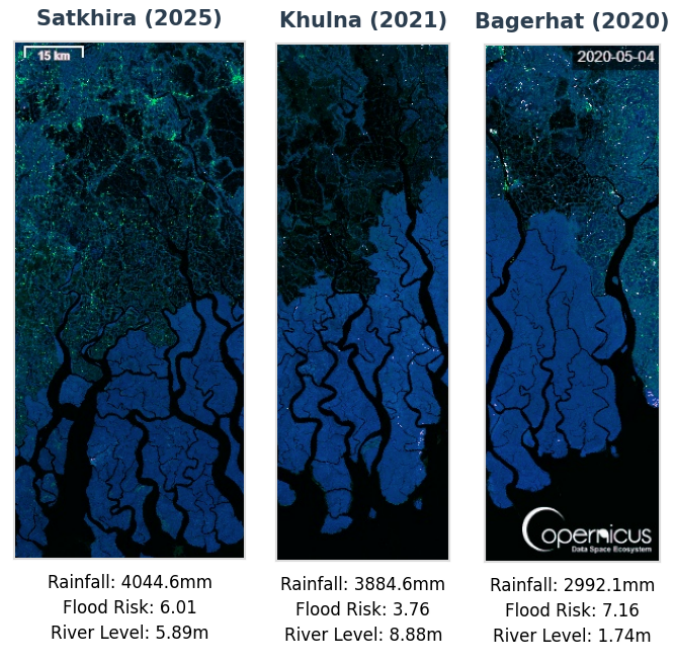


Fig. 3. Visualization of the preprocessed data.

C. Data Augmentation

We applied various data augmentation techniques during training to ensure that our model generalizes effectively and does not simply memorize the training images. The input images were transformed randomly by these:

- **Random Rotation:** Images were rotated by as much as 20 degrees to account for variations in satellite orientation.
- **Horizontal Flip:** Random left-to-right flips were applied to learn the model that flood features do not depend on direction.
- **Color Jitter:** Minor adjustments were made to the brightness and contrast settings to replicate different sensor conditions.

D. Proposed Architecture

To handle the multimodal data, we developed a dual-stream neural network architecture called "SARA-Net" (Sundarbans Adaptive Risk Assessment Network). There are three primary parts to the architecture:

- 1) **Visual Feature Extractor:** The image branch uses a ResNet-18 backbone, which is a Convolutional Neural Network (CNN) designed to extract high-level spatial features from SAR images, such as water textures and land boundaries.
- 2) **Climate Feature Extractor:** The text branch uses a Multi-Layer Perceptron (MLP) model. This fully connected network processes numerical rainfall and river data to identify statistical drivers of floods.
- 3) **Fusion Head:** The features from both branches are combined into a single vector. This unified representation is then processed by a final classification block comprising dense layers and a Sigmoid activation function. As a result, it produces a probability score ranging from 0 (indicating Safe) to 1 (indicating High Risk).

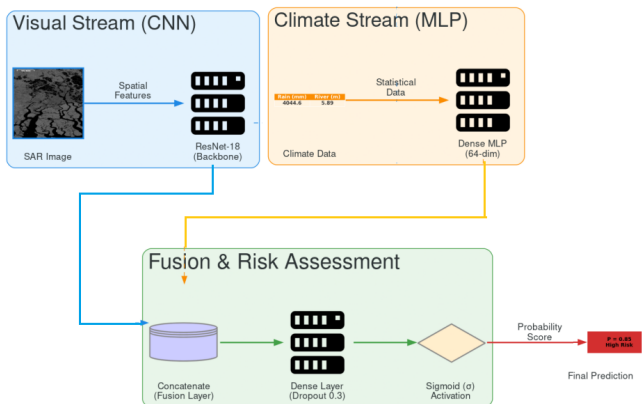


Fig. 4. The detailed architecture of the SARA-Net model.

E. X-AI Implementation

To address the "black box" nature of deep learning, we integrated an Explainable AI (X-AI) module using the Layer-CAM technique. We set up forward and backward hooks on the final convolutional layer of the ResNet-18 backbone. This approach allows us to capture the gradients that flow back from the predictions. By multiplying these gradients with the feature maps, we create a localization map that highlights the specific pixels in the SAR image that contributed most to the "High Risk" prediction.

F. Modality Dropout Strategy

To force the model to learn robustly from both data sources, we implemented a "Modality Dropout" method during training. In 10% of the training batches, we randomly removed either the image or the text input. This regularization technique reduces the model's reliance on any single data modality.

G. Experimental Setup

The model was implemented using the PyTorch framework and executed in a Kaggle T4 GPU environment. We utilized the Binary Cross-Entropy (BCE) loss function to assess the error between the predicted values and the actual labels. The network was optimized with the Adam optimizer, set to a learning rate of 0.00005. We trained the model for 50 epochs to ensure convergence.

RESULTS AND DISCUSSION

To evaluate the practical performance of the SARA-Net model, we utilized 775 distinct samples from the districts of Satkhira, Khulna, and Bagerhat. The model achieved a final accuracy of 83.71%, demonstrating its capability to differentiate between "Safe" and "High Risk" flood scenarios. We also analyzed the Receiver Operating Characteristic (ROC) curve, where the model obtained an Area Under the Curve (AUC) score of 0.9349 (see Fig. 5b). A high AUC score indicates that the model can effectively detect floods while minimizing false alarms. This balanced performance across both categories is further supported by the Confusion Matrix (see Fig. 5a).

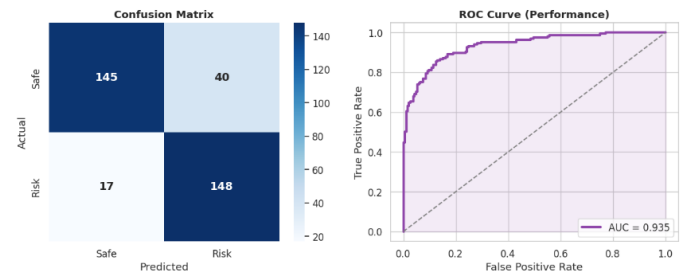


Fig. 5. Model Performance: (a) Confusion Matrix, (b) ROC Curve.

We used selective input removal in a feature importance test to determine which data source was responsible for these predictions. When the SAR Image input was removed, the

model’s error increased significantly by +0.4139. In contrast, removing the Climate Data only increased the error by +0.0390. This disparity demonstrates that the visual features in SAR images, such as water texture and land boundaries, are the primary drivers of the model’s performance. At the same time, the climate data provides helpful but secondary context.

We applied the trained model to historical data from 2014 to 2025 in order to reconstruct a long-term risk profile. Figure 6 illustrates the trend in flood vulnerability across the three districts over these 12 years. The analysis reveals a clear upward trend in risk, particularly in the Satkhira district (indicated by the red line). The model accurately identified 2020, the year of Cyclone Amphan [27], and 2022 as peak vulnerability years, during which the risk score exceeded the 80% threshold. In comparison, Bagerhat (represented by the green line) displayed lower volatility, indicating varying levels of climate stress across the region.

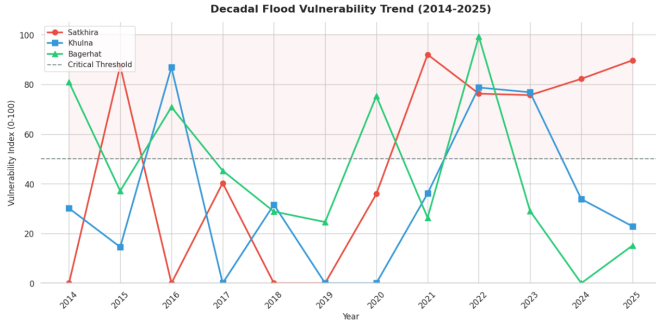


Fig. 6. Flood Vulnerability Trend (2014-2025).

TABLE I
HYDROLOGICAL COMPARISON OF BAGERHAT DISTRICT (2022 vs. 2025).

Year	Scenario	Rainfall (mm)	River Level (m)	Impact Score
2022	Worst Case	4506.8	9.79	5.26
2025	Normal	1451.3	4.15	0.26

The Bagerhat district was analyzed across two years (Fig. 7) and cross-referenced the results with actual environmental data from our Dataset 2 to confirm the model’s understanding of various flood stages. The “Worst Case” scenario in 2022 is shown in the left figure. This year was characterized by significant hydrological stress, with annual rainfall reaching 4506.8 mm and river levels rising to 9.79 m, resulting in a critical flood effect score of 5.26 (see Table I). The heatmap provides a clear picture of intense, widespread activation, confirming significant flooding. In contrast, the right image depicts 2025, characterized by stability (Rainfall: 1451.3 mm, Impact Score: 0.26). During this period, the model’s focus is restricted to the riverbanks. This comparison shows that the model effectively distinguishes between severe flooding driven by climatic extremes and normal hydrological conditions.

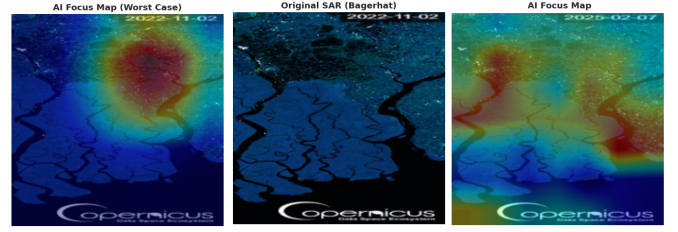


Fig. 7. AI Visual Analysis of Bagerhat Zone.

Finally, we performed a targeted analysis of Satkhira, which our trend data identified as the most critical region. Figure 8 presents the AI’s focus map for Satkhira. The intense red heatmap indicates a significantly larger affected area than in other districts. This visual evidence confirms that Satkhira experiences the most extensive and persistent flooding, supporting the statistical findings that it is the most vulnerable area in the Sundarbans.

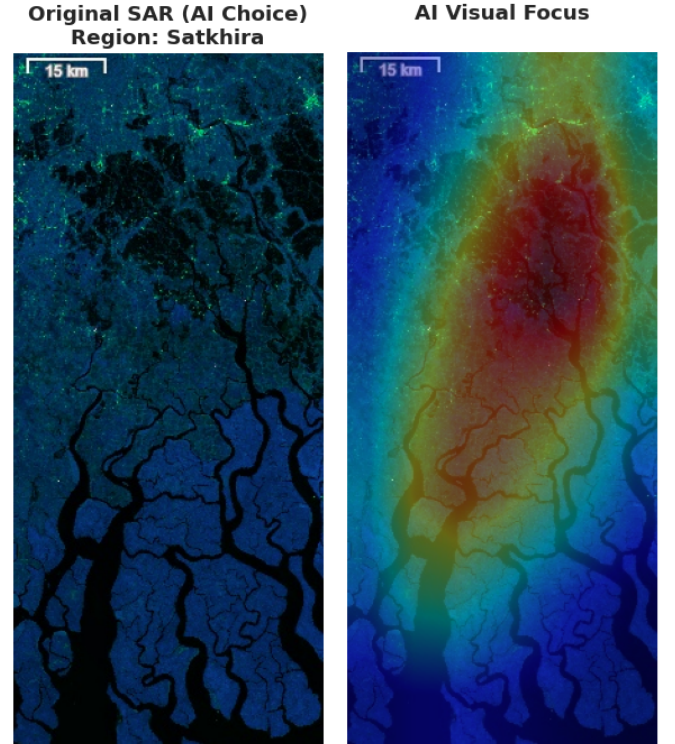


Fig. 8. Vulnerability Analysis of Satkhira Zone

CONCLUSION

In this paper, a deep learning model, SARA-Net, was introduced to assess flood risks in the vulnerable Sundarbans region. By integrating satellite radar imagery with statistical climate data, our dual-stream architecture effectively identified high-risk areas with 83.71% accuracy and an AUC score of 0.9349. The results showed that combining visual data with climate records greatly enhances prediction reliability compared to using either source independently. Additionally, our long-term analysis accurately highlighted significant historical

events, including Cyclone Amphan in 2020, and determined that Satkhira was the district most at risk of severe flooding.

The study has some limitations despite these promising findings. First, the model relies on the Sundarbans' specific geography, so it may need to be further developed to work effectively in other regions with different landscapes. Second, processing two data streams simultaneously is computationally intensive. Our model currently requires powerful hardware, such as the T4 GPU used in our experiments, and is challenging to run on standard computers or portable devices. However, the resolution of public satellite data restricts our ability to detect very small or localized flooding events in densely populated urban areas.

Future work shall focus on addressing these challenges to improve the system's usability for real-world applications. We plan to expand the dataset to include additional coastal regions, which will help the model generalize better to new areas. Additionally, we aim to optimize the network architecture to make it more lightweight, enabling it to run on edge devices for faster alerts. Ultimately, we intend to connect the model to real-time satellite feeds, transforming this research tool into a live disaster monitoring system that can provide timely warnings to local communities.

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